Filename: 1-7-1-common-network-architecture.md
Domain: System and Network Architecture Concepts in Security Operations
Episode: Common Network Architecture
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Common Network Architecture

Objectives

- On-Prem
- Cloud
- Hybrid
    - Wired and Wireless connections
    - Public and Private networks
    - Cloud and On-Prem resources
- Network Segementation
    - Logically and/or Physically segmented
    - Limits attacker's ability to laterally move
    - Access to critical systems and data are also limited
- Zero Trust
    - Never Trust. Always verify.
        - Request and connections are NEVER trusted
        - Applies to External AND Internal sources and locations
            - Employs a lot of Monitoring and Analysis
                - IDS/IPS
                - User Entity Behavior Analytics
                - Traffic Analysis
- Secure Access Secure Edge (SASE)
    - Provides secure access to network resources from anywhere
        - Combines
        - SD-WAN
        - SaaS
        - CASBs
        - Zero Trust Network Access
- Software-Defined Networking (SDN)
    - Traditional Networking
        - Control Plane
        - Data Plane
            - Each network device must be configured by the admin
                - These devices(routers/switches/etc) handle both the Control and Data Planes to make decisions about where traffic should go, and then actually send the traffic
    - SDN
        - Does all that, but in a different way
        - Centralized software-based controller
            - Single point of control for the entire network
                - Admin defines policies and rules for how network data should flow
                - SDN makes it happen
        - Toy Cars analogy

- You have a bunch of toy cars that you wish to move around
- You have a friend that knows everything about those cars and can communicate with their GPS
- You tell your friend how you want the cars to move
    - Sedans go North
    - Coupes go South
    - Trucks go East
    - Vans go West
- Your friend then communicates with the cars to get them where they need to go
    - "Updates their GPS"